Lean formalization of an Assumptions of Physics technical brief: log

This log records which statements of the technical brief in this directory were checked in Lean, where the formal version differs from the brief, and any problems found in the brief. The brief is available at `https://assumptionsofphysics.org/resources/briefs`.

Setup

- Lean `v4.36.0-rc1`; Mathlib at commit `2cabef97`; physlib (`master`) and TauCeti (commit `eb4dd24`, on the same Mathlib) as further dependencies.
- Build: `lake build` in the project root.
- Every file is checked with `#print axioms` on its main results. A result is recorded here only if it depends on no axioms beyond `propext`, `Classical.choice` and `Quot.sound`, and so has no `sorry`.
- Imports are kept to the Mathlib modules actually needed.
- Material that belongs in Mathlib is kept in `ForMathlib/`, in Mathlib's directory layout (library `ForMathlib`), and imported by the brief files.

1 Unphysicality of Hilbert spaces for quantum mechanics

Files.

- File: `briefs/HilbertUnphysicality/HilbertUnphysicality.lean`
- Library: `HilbertUnphysicality`
- Imports: `InconsistentConvergence` and three new `ForMathlib` files:
  `ForMathlib/Analysis/SpecialFunctions/Gaussian/Erf.lean` (the error function),
  `ForMathlib/Analysis/InnerProductSpace/l2Space.lean` (separable Hilbert spaces are $\ell^2(\mathbb{N})$),
  `ForMathlib/MeasureTheory/Function/L2Space.lean` (L^2 is infinite-dimensional)

Setting. Theorem 1 says that the Hilbert-space formulation QSTO:H is inconsistent with three physical conditions. The conditions are informal; what is checked is the mathematics behind each of the three arguments.

Results.

Brief	Lean
[B]COBS	
unbounded ⇒ not continuous	`not_continuous_of_unbounded`
oscillator example	`numberExp_not_continuousAt` (from the inconsistent-convergence brief)
[B]WDQ	
Hellinger–Toeplitz	`bounded_of_isSymmetric`, `not_isSymmetric_of_unbounded`
erf, $\lvert\operatorname{erf}\rvert < 1$	`Real.erf`, `Real.abs_erf_lt_one`, `Real.tendsto_erf_atTop`
$y(x) = \tan(\frac{\pi}{2}\operatorname{erf} x)$ is a change of coordinates	`frameChange`, `strictMono_frameChange`
eq. (2), first line, integrated	`arctan_frameChange`: $\arctan(y)/\pi = \operatorname{erf}(x)/2$
eq. (2), second line	`cauchyDensity_frameChange_mul_deriv`: $\lvert\phi(y(x))\rvert^2\, y'(x) = \lvert\psi(x)\rvert^2$
ψ normalized, in the domain of X	`integral_gaussDensity`, `integrable_sq_mul_gaussDensity`
ϕ normalized, not in the domain of Y	`integral_cauchyDensity`, `not_integrable_sq_mul_cauchyDensity`
[B]STDIST	
separable Hilbert spaces are isomorphic	`nonempty_linearIsometryEquiv_of_separable`
$L^2(\mathbb{R}^n)$ is separable and infinite-dimensional	`L2.separableSpace_euclidean`, `L2.not_finiteDimensional_euclidean`
$L^2(\mathbb{R}^{n+1}) \cong L^2(\mathbb{R}^{m+1})$	`L2_euclidean_iso`

Differences from the brief.

- The wave functions are handled through their densities $|\psi|^2 = e^{-x^2}/\sqrt{\pi}$ and $|\phi|^2 = 1/(\pi(1+y^2))$. "In the domain of the position operator" is read as $\int x^2|\psi|^2 < \infty$, i.e. $x\psi \in L^2$.
- The frame change is checked as a strictly increasing coordinate change satisfying eq. (2) in both forms. The unitary U between the two L^2 spaces and the change of variables in the integral are not formalized.
- Mathlib has no error function; `Real.erf` is defined in `ForMathlib`.
- [B]STDIST is proved for $n, m \geq 1$ through the classification of separable infinite-dimensional Hilbert spaces, which is also new in `ForMathlib`.
- The additional remarks of Section 4 are not formalized.

Problems found in the brief.

- The brief says that $\phi(y)$ "goes to zero at infinity like $1/y^2$". It is $|\phi|^2$ that decays like $1/y^2$; ϕ itself decays like $1/|y|$. The conclusion is unaffected: $y\,\phi(y) \notin L^2$.
- "$L^2(\mathbb{R}^n)$ is a separable Hilbert space for any n" is true, but the isomorphism argument also needs infinite dimension, which fails for $n = 0$.

- QSTO:H allows any separable Hilbert space, finite-dimensional ones included. In finite dimension every self-adjoint operator is bounded and defined everywhere, so [B]COBS and [B]WDQ hold, and Theorem 1 is false as stated. It needs $\mathcal{H}$ to be infinite-dimensional, or equivalently some observable to be unbounded.

- The [B]STDIST argument holds only if the representation of a system is the Hilbert space, with what QSTO:H builds from it alone, up to unitary equivalence. If the representation also includes the positions $X_1, \ldots, X_n$ and the momenta $P_1, \ldots, P_n$, then n is part of the data and [B]STDIST holds trivially. The brief does not say which is meant. Under the first reading [B]STDIST fails even in finite dimension: the spin of an electron and the polarization of a photon are both represented on $\mathbb{C}^2$.